Measured hourly business processes

Each fact is an Hourly Periodic Snapshot of an operating interval.

FactHourlyDemand

Specification	Definition
Grain	One selected authority and one UTC hour-ending instant with its demand, forecast and total net generation.
Primary key	DemandKey
Business uniqueness	(AuthorityKey, UTCHourEnd)
Measures	ActualDemandMW, ForecastDemandMW, TotalGenerationMW
Parents	DimBalancingAuthority, DimDate, DimHour, DimSourceSnapshot
Revision behavior	Latest published values per business grain. Revisions update the same fact row; preserve raw snapshots for audit.

FactHourlyGeneration

Specification	Definition
Grain	One selected authority, UTC hour-ending instant and original fuel category with a numeric reported value.
Primary key	GenerationKey
Business uniqueness	(AuthorityKey, UTCHourEnd, EnergySourceKey)
Measures	GenerationMW
Parents	DimBalancingAuthority, DimDate, DimHour, DimEnergySource, DimSourceSnapshot
Revision behavior	Latest published values per business grain. Revisions update the same fact row; preserve raw snapshots for audit.

The demand fact keeps a row even when one measure is missing, preserving the observed-hour population. The generation fact stores numeric reported fuel observations, including zero and negative values. Empty cells and absent categories remain diagnosable from frozen source rows and schemas.

Descriptions that make facts comparable

Parent keys are unique. A fact foreign key references exactly one dimension row.

Dimension / natural key	Attributes and hierarchy	History
DimDate / CalendarDate	CalendarYear, QuarterNumber, MonthNumber, ISOWeekYear, ISOWeekNumber, WeekdayNumber, Season. Date to month/quarter/year; separate ISO week path.	Type 0. Calendar attributes are deterministic.
DimHour / HourEnding	HourLabel. One clock-ending hour 00-23. No unique date identity in this dimension.	Type 0. A clock label does not identify a unique instant.
DimBalancingAuthority / AuthorityCode	AuthorityName, RegionCode, RegionName. Authority to region.	Type 1 for current reference labels. Historical renaming is not reconstructed from a current reference workbook.
DimEnergySource / SourceColumn	SourceLabel, SourceGroup. Original category to reviewed group.	Type 0 meaning. A newly split source category receives a new member. Keep legacy combined categories explicit.
DimSourceSnapshot / FileName + SHA256	SourceURL, RetrievedAtUTC, SourceRowCount. Immutable file-version lineage.	Immutable. A different file hash creates a new member; unchanged bytes reuse the member.

Conformed dimensions

Date, Hour, BalancingAuthority and SourceSnapshot have shared keys and definitions across both facts. EnergySource applies to generation. Region remains an attribute of the authority dimension, keeping the model denormalized.

SCD choice

The baseline provides current reference descriptions and timestamped observations. Type 1 updates descriptive spelling/name corrections. Type 2 is not required for the selected pilot because an authoritative history of attribute changes was not acquired. If historical authority membership becomes necessary, it requires dated reference evidence and a reviewed Type 2 extension.

Keys, additive behavior and nulls

The physical atlas and Excel dictionary are generated from the same table and column definition.

Element	Rule
MW measures	Add only across disjoint components at the same hour. Use mean and maximum over time. Convert one-hour observations to MWh for period energy.
Forecast error	Compute from paired observations. MAE, bias and WAPE use eligible pairs and must be recalculated at each grouping.
Time keys	DateKey is the source operating day. HourKey is the local end-clock label. The UTC timestamp establishes the unique observation instant.
Lineage	SourceSnapshotKey identifies file bytes, URL and extraction time. SourceRowNumber points to the one-based original data record.
Unknown / N/A	Required time and identity keys are resolved or held. Blank measures are null. Published Other/unknown fuel categories are valid members. No invented key is used to hide an orphan fact.
Fact-to-fact analysis	FactHourlyDemand is unique on authority and UTC hour. FactHourlyGeneration adds energy source to that grain. Aggregate generation to authority and UTC hour before joining to demand. For a group comparison, aggregate each fact independently to the same dimensions and then combine. One shared authority-hour with eight fuel categories still has one demand observation.

Keep immutable raw extracts and their hashes. A changed bulk file creates a new DimSourceSnapshot member. A later load updates measures at the same business grain and preserves fact surrogate IDs. Replacement of a source partition must also remove fuel facts for values withdrawn by the publisher, or mark them absent through a reviewed policy. A latest-value fact cannot reproduce an old publication without its archived source snapshot.